Translation of Aerial Image Into Digital Map via Discriminative Segmentation and Creative Generation

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***Abstract*— Automatic translation of aerial images into digital maps is an important and challenging task which is widely used in practical applications. Most of the existing works view it either as a creative image-to-image translation problem or a discriminative semantic segmentation problem. However, we notice that human annotators need to extract and understand the information in aerial images first and then translate them to online maps in a creative way, which helps them draw accurate and visually appealing online maps. In this article, we propose an end-to-end online map generation method that combines a discriminative module with a creative module based on this observation to mimic human behavior. Specifically, we first utilize a semantic segmentation module to obtain a rough aerial map, in which each region is labeled with its category, and then further improve its quality with a creative module. To train a robust network that generalizes well to unfamiliar regions, we also collect a large aerial image dataset for online map generation (AIDOMG). AIDOMG consists of 40 087 pairs of aerial images and corresponding online maps collected from nine regions of six continents. We conduct extensive experiments to verify the superiority of the new design that combines discrimination and creativity and experimental results show that the performance of the proposed method significantly outperforms baseline methods.**

***Index Terms*— Aerial image, generative adversarial networks (GAN), map generation, semantic segmentation.**

I. Introduction

ONLINE maps service plays an important role in modern life. It not only provides convenience for individuals but also facilitates the development of commercial companies, such as ride-sharing companies and food delivery companies. However, widely used online maps are usually rendered from vector data, whose collection and update are both time consuming and expensive [1], [2]. These weaknesses limit its application.

Compared with map vector data, aerial images are easier and cheaper to obtain. They naturally contain rich information of the ground surface and can be utilized for map generation. The collection and update of them can be done with airplanes, satellites, and drones, which is not influenced by ground conditions and have better adaptability. More importantly, they can be acquired timely even under extreme situations, such as earthquakes, floods, mudslides, and other natural disasters, where timely updated online maps can provide rescuers with important information for saving lives.



Fig. 1. Online maps of different qualities. (a) Original aerial image. (b) Online map with accurate road location and good visual quality. (c) Online map generated by image-to-image translation methods, which visually looks like a map, but the road location is not accurate. (d) Online map generated by semantic segmentation methods, the road position roughly corresponds to aerial image, but the visual quality is not like a map.

Online map generation task is to automatically generate online maps of the corresponding area based on easily accessible information, such as aerial images. Currently, there are two branches of approaches that can convert aerial images to online maps. One is based on image-to-image translation, the other is based on semantic segmentation. Image-to-image translation methods [3]–[6] view aerial images and corresponding maps as images from two different domains and aim to translate aerial images into the target online map images directly. As shown in Fig. 1(c), online maps generated in this way have realistic visual quality and image style but suffer from semantic incorrectness. Semantic segmentation methods [7]–[11] regard this task as an image segmentation problem, i.e., classify every pixel in aerial images to a class label. As shown in Fig. 1(d), they can generate relatively accurate online maps, but results are not of high visual quality.

In this article, we aim to generate accurate online maps with high visual quality. To this end, we notice the fact that humans need to extract and understand the information in aerial images first and then draw a realistic online map image in a creative way [12]–[14]. Based on this observation, we propose an end-to-end online map generation method with two modules, including a discriminative module and a creative module. They are based on semantic segmentation and image-to-image translation, respectively. First, we use a discriminative module

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to extract ground information from aerial images. It is able to understand images and has pixel-level accuracy in recognizing semantic labels. Second, we use a creative module to learn the prior knowledge of real online maps and generate final online map images. It refines details and styles of the generated online maps, making them look real with satisfactory visual quality. Our method embraces the strength of semantic segmentation methods and image-to-image translation methods and uses the two modules in close collaboration during training. In this way, our pipeline is able to generate accurate and realistic online maps from aerial images.

To verify the effectiveness of our method and make up for the deficiency of existing datasets for online maps generation, we collect a dataset from real-world scenarios. It has high-quality samples from nine regions of six continents with varying landform features and provides semantic segmentation labels, i.e., each sample consists of an aerial image, corresponding online map image, and semantic segmentation labels.

In summary, our contributions are twofold.[1]

1) We propose an end-to-end pipeline with two modules for generating online maps from aerial images. It closely combines stylistic and semantic features, and the online maps it generates show advantages in different metrics.
2) We collect a dataset for online maps generation that provides aerial images, corresponding online maps, and semantic segmentation labels. It has high-quality samples with varying landform features, which can serve as a standard benchmark for online maps generation tasks.

The remainder of this article is organized as follows. We review related work in Section II. We picture the main design and describe details of our method in Section III. In Section IV, we introduce a large-scale collected dataset named aerial images dataset for online map generation (AIDOMG). We provide experimental results and comparisons in Section V. Finally, the conclusion is drawn in Section VI.

II. Related Work

In this section, we first review the development of image-to-image translation methods and semantic segmentation methods. Then, we introduce techniques related to the task of online maps generation. Finally, datasets for online maps generation are introduced.

A. Image-to-Image Translation Methods

Image-to-image translation (img2img) is the task of transforming images from one domain to another. Isola *et al*. [3] proposed an image-to-image translation method using conditional generative adversarial networks (cGANs) [15], which has been used for many translation tasks, e.g., labels to the street scene, day to night, edges to photograph, aerial to map. Many methods after that have shown impressive results in different situations, e.g., [16] is able to generate high-resolution images with good quality, [17] can be trained with unpaired dataset, [18]–[20] can translate images from multiple domains, and [21] performs better in content-rich scenes. Image-to-image translation methods are also involved in many specific applications, e.g., portrait editing [22], image colorization [23], style transfer [24], and map generation [4], [5]. Image-to-image translation methods can easily learn the mapping between images from different domains, but it does not explicitly understand the content in images.

B. Semantic Segmentation Methods

Semantic segmentation is one of the key problems in computer vision, and it has been widely used in aerial image analysis. Long *et al*. [25] proposed an end-to-end fully convolutional network to output 2-D prediction images. Then, methods after that performed better on contextual information using multiscale attention [7], [26]–[33]. DeepLab family [7], [34]–[36] is currently one of the best semantic segmentation methods, which uses atrous spatial pyramid pooling (ASPP) structure to fuse multiscale information for segmentation, and uses decoder module to refine segmentation by fusing low-level and high-level features. Based on these methods, semantic segmentation is used to analyze aerial images for various applications, e.g., land use [10], [37], building extraction [11], vegetation distribution [38], and vegetation growth [39].

C. Online Map Generation Based on Aerial Images

1) Manual-Based Methods: Traditional digital maps usually rely on manual painting [1], [2]. The staves first conduct on-site surveying and mapping on the spot and record vector data. Then, they draw RGB map images based on the vector data and split them according to certain specifications to form digital maps. This kind of manually drawn map has high accuracy and quality, but it takes much time to make, and the update cycle is long (usually several years). Moreover, it can hardly cope with sudden disasters, such as earthquakes, floods, and mudslides, which may lead to changes in the topography.

2) Semantic Segmentation Methods: Semantic segmentation can extract specific information from aerial images, which can be useful for automatic map generation, but it does not directly generate online maps. Semantic segmentation methods e.g., [7] and [30] may be helpful for aerial images analysis, e.g., we can identify a specific type of ground target from aerial images using semantic segmentation methods [10], [37], [40]–[43]. However, semantic segmentation networks usually pay no attention to image style and detailed features of output results. For example, networks may predict false negatives on edge of the object when input images are blurry, making the segmentation result of road or river looks not smooth enough. Furthermore, semantic segmentation networks lack creativity, their solution space is usually limited by discrete categories corresponding to the dataset.

3) Vector Extraction Methods: In addition, there are some methods that try to obtain vector data from aerial imagery, such as building vector extraction methods [44]–[46] and road vector extraction methods [47], [48]. Building vector extraction methods usually first predict the raster masks

[1]Our dataset and code will be publicly released at "https://github.com/ying-fu/Aerial-Map-Translation" soon. This article must be cited when the dataset or code is used for research purposes.

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of buildings using semantic segmentation pipeline and then process them into standard vectorized polygons [44]–[46]. Road vector extraction methods, on the other hand, try to draw vectorized road graphs step by step from start points [47], [48]. They usually focus on a single ground element class, such as building or road. These methods can provide vector information to facilitate the use of geographic information systems (GISs). If these methods are used to generate online maps, it is necessary to handle different categories of ground elements separately, handle possible conflicts, and then set up the rendering method to render online maps. This will increase the complexity of these methods.

4) GAN-Based Methods: Recently, the rapid development of deep learning has made automatically generate high-quality online maps from aerial images become a possible research objective. Mainstream methods regard this task as an image generation task, some of them [3], [16], [17] can convert aerial images directly into online maps. Considering that the characteristics of online map conversion and ordinary image conversion are different, [4]–[6] attempted to adjust the model and loss function according to the characteristics of the online map to improve the quality of the generated online map. GAN-based methods can continuously optimize the output in the confrontation between the generator G and the discriminator D. The discriminator D can judge whether an image is real based on high-order features and learn a loss to penalize differences between outputs and targets. It makes the generator G continuously improve the visual effect of outputs, and finally, makes the generator have good creativity. However, GAN is unable to understand the contents in images explicitly and usually lacks discriminative ability.

5) Combination Method With Creative and Discrimination: Based on the above-mentioned two types of methods, we proposed a novel method combining them together to further handle their limitations, in which we effectively take creativity and discrimination into account. We use a discriminative elements-extracting module based on semantic segmentation to explicitly extract semantic information of aerial images and a creative map generation module based on image-to-image translation to generate online map images.

D. Datasets

Current aerial image datasets with pixelwise labels can be divided into two categories. One is the aerial image and online map image dataset, e.g., the "map" dataset used in pix2pix (PP dataset) [3]. It consists of paired aerial images and online map images for map generation tasks. Currently, this type of dataset is relatively small and usually focuses on a single-urban scene. The other is a dataset for land cover classification, which labels the land use of each pixel, e.g., [49], [50], and consists of pairs of aerial images and land information images. It usually contains more types of land information, such as urban land, garden land, irrigated land, and usually pays no attention to image style. They are commonly used for semantic segmentation tasks of aerial images. In addition, because of the high labeling cost of pixelwise labels, more aerial image datasets are annotated with objectwise labels, e.g., DOTA [51], VHR-10 [52]–[54], HRRSD [55], and DIOR [56].

In this article, we propose the AIDOMG. It samples from nine areas and has a large number of images, covering various ground situations. In addition, we prepare corresponding semantic labels for each aerial image.

III. Method

In this section, we first formulate the problem and introduce the motivation, and then we describe our pipeline with two modules. Finally, learning details are provided.

A. Formulation and Motivation

Though online map generation can be regarded as an image-to-image translation task, it is more challenging because it needs not only to transform images to the style of traditional online maps but also to keep accurate ground information in aerial images, which image-to-image translation do not take into account. To address this, we get inspiration from manual map generation. We notice that humans need to extract and understand the information in aerial images first and then translate them to another style in a creative way [12]–[14].

Therefore, we propose a novel online map generation method with two modules. The first one is the discriminative module which can discriminate semantic categories of each pixel in aerial images. And the second one is a creative module that can learn styles and features from real-online map images and restore these styles and features as much as possible in the outputs.

B. Pipeline

Our pipeline is designed to generate accurate online maps from aerial images, as shown in Fig. 2, it consists of two modules, discriminative module and creative module. To ensure accuracy, we employ a discriminative module to extract semantic information from aerial images. And the creative module is designed to translate the ground semantic information to online maps in a creative way.

1) Discriminative Module: High-quality generated maps need to be accurate to display a realistic view of the world. Existing GAN-based map generation approaches [4], [5] focus on the visual quality of the generated image and ignore the semantic information, which leads to inaccurate map generation. To generate accurate online maps, we use a discriminative module to explicitly extract semantic information from aerial images. Our discriminative module follows the semantic segmentation method DeepLabv3+ [7] for its stable performance, generality, and ease of training. Specifically, we use Xception [57] as the main part of the encoder to extract feature maps. The stride of the encoder is set to 16, which denotes the size of feature maps is 1/16 of the original images. After that, we use the ASPP module to process feature maps, which extracts multiscale features by applying different-scale cavity convolutions in parallel. Then, we concatenate high-level features obtained by the encoder with low-level features from previous layers. Considering that the importance of low-level features is usually weaker than high-level features, we use 1×1 convolution on low-level features to reduce the number of channels. Finally, we use several convolutional

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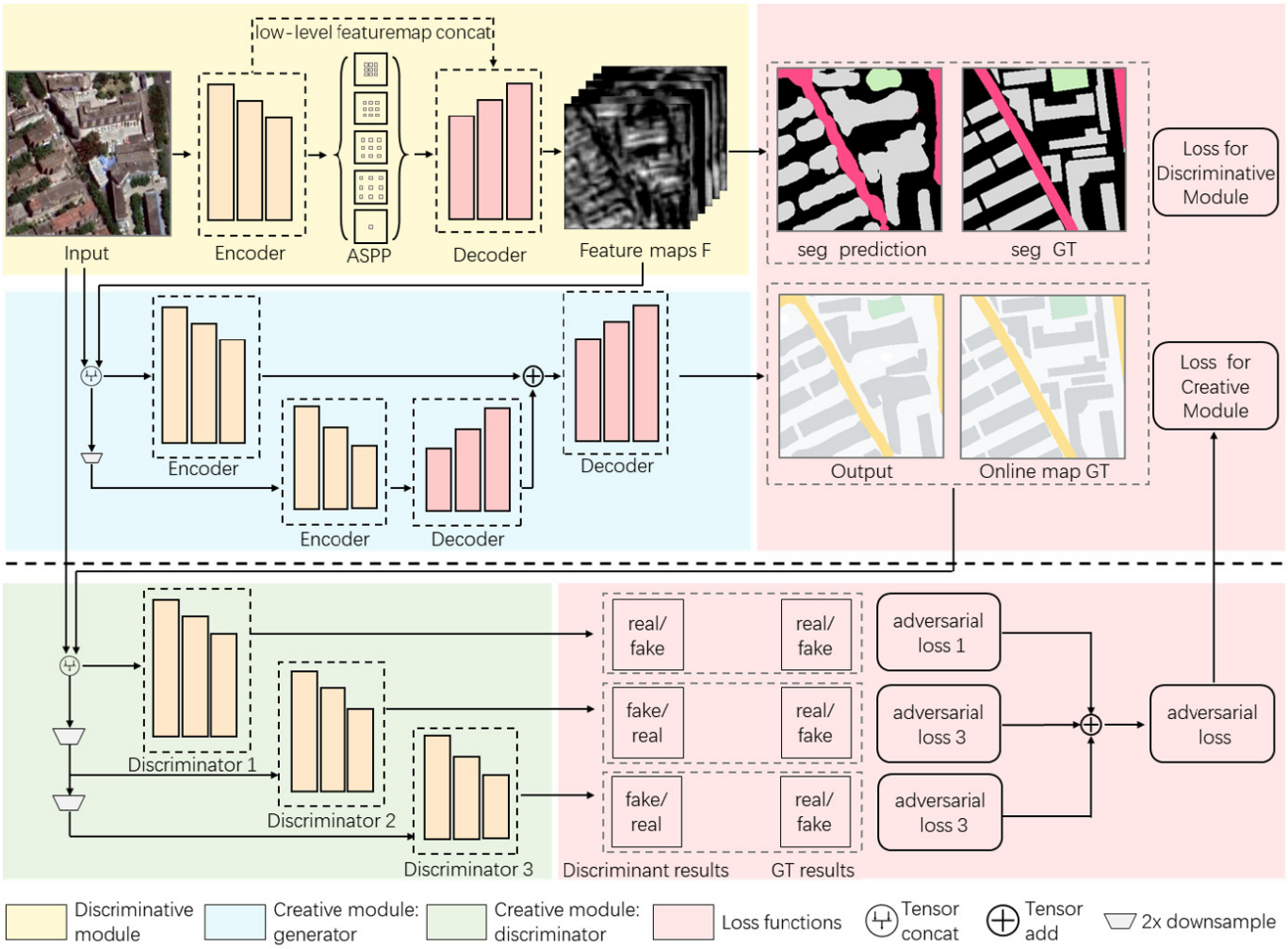


Fig. 2. Overall pipeline. It consists of a discriminative module and a creative module. Discriminative module extracts semantic information from input aerial images pixel by pixel and creative module using semantic information to generate output online map images.

layers to predict semantic segmentation results. In the decoding process, there are two 4× bilinear upsampling layers to ensure that the size of the final prediction result is consistent with the input image.

2) Creative Module: The discriminative module is able to extract accurate pixelwise semantic information from aerial images, however, the visual quality of its segmentation results is inferior compared with real online maps for lacking realistic details. To improve the visual quality of generated maps and make them easy to understand, we employ a creative module based on cGAN [15]. It takes aerial images and semantic information as input. The structure of generator is based on U-Net [3], [17], [58], and we use a coarse-to-fine structure to refine details [16]. Specifically, our generator consists of two subnetworks, i.e., a global generator and a local generator. The global generator focuses on the whole performance of image, it receives the 2× downsampled low-resolution aerial images and the semantic information from the discriminative module as input to generate low-resolution output maps. The local generator focuses on the details, and it receives the original resolution aerial images and semantic information as input and fuses the results from the global generator using pixelwise addition in the middle layer. We use feature maps extracted by the discriminative module as the carrier of semantic information, and concatenate them with the aerial image on the channel dimension as the input to generators, i.e., the shape of the input is $[B, C_A+C_S, H, W]$, where B denotes the batch size, C_A and C_S are the number of channels of aerial images and the semantic feature maps, respectively, and H and W are the height and width of the aerial images, respectively. We adopt multiscale discriminator [16] to enhance the ability to differentiate real and generated maps at different scales. Specifically, we have three discriminators, D_1, D_2, and D_3, with the same structure and they receive images of different sizes. Each discriminator consists of three downsampling convolutional layers with stride 2 and two convolutional layers with stride 1 and finally outputs a one-channel confidence map. D_1 receives aerial images and online maps with original size, and D_2 and D_3 receive 2× and 4× downsampled aerial images and online maps.

C. Loss Function

In order to train the model effectively, our loss functions can be divided into two parts, i.e., loss for the discriminative

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module and loss for the creative module. The loss function is expressed as

$$\mathcal{L} = \lambda \mathcal{L}_{\mathrm{DM}} + \mathcal{L}_{\mathrm{CM}} \tag{1}$$

where $\mathcal{L}_{\mathrm{DM}}$ denotes loss for discriminative module and $\mathcal{L}_{\mathrm{CM}}$ denotes loss for creative module. We set λ to 0.5 by default according to experiments. The specific experimental results are described in the ablation study.

1) Loss for Discriminative Module: The discriminative module is trained to understand aerial images pixelwisely, we follow [7], [25], and [33] to use pixelwise cross entropy loss for training. To better adopt hard samples and category imbalance, we use focal loss [59], a common improvement of cross entropy loss. The loss function is

$$\mathcal{L}_{\mathrm{foc}} = \begin{cases} -\alpha_t (1-p)^{\gamma} \log(p), & \text{if } y = t \\ -\alpha_t \ p^{\gamma} \log(1-p), & \text{otherwise} \end{cases} \tag{2}$$

where α is balance variable, γ is tunable focusing parameter, $y \in \{0, 1, \ldots, c-1\}$ is the ground truth class, and $p \in [0, 1]$ is the estimated probability for class with label $y = t$. We set α_t as the reciprocal of the area of category t to balance the impact of differences in the number of samples from different categories. We set γ to 2 by default.

In addition, we adopt Jaccard loss [60] to improve the mean intersection over union results

$$\mathcal{L}_{\mathrm{Jaccard}} = 1 - \frac{|\{y^* = c\} \cap \{\tilde{y} = c\}|}{|\{y^* = c\} \cup \{\tilde{y} = c\}|} \tag{3}$$

where y^* is vector of ground truth labels and $\tilde{y}$ is vector of predicted labels. Since Jaccard loss is difficult to derivate, we use Lovász loss [60] $\mathcal{L}_{\mathrm{lov}}$, which is a convex surrogate to the Jaccard loss.

Together, our loss for discriminative module is expressed as

$$\mathcal{L}_{\mathrm{DM}} = \mathcal{L}_{\mathrm{foc}} + \mathcal{L}_{\mathrm{lov}}. \tag{4}$$

2) Loss for Creative Module: The creative module is trained to generate online maps with more realistic visual quality and details. To this end, we adopt content loss, adversarial loss, perceptual loss, and feature-matching loss for training.

The content loss [3], [16] $\mathcal{L}_{\mathrm{con}}$ is for improving the similarity between generated map and real map pixelwisely. We use mean absolute error to constrain our model. The loss function is

$$\mathcal{L}_{\mathrm{con}} = \frac{1}{n} \sum_{i=1}^{n} |G(a)_i - m_i| \tag{5}$$

where n is the total number of pixels, G is generator, a denotes aerial images, and m denotes map images.

The adversarial loss [61] $\mathcal{L}_{\mathrm{adv}}$ is a learned loss that can make details of generated maps more realistic. The loss function is

$$\begin{aligned} \mathcal{L}_{\mathrm{adv}} &= \min_G \max_D V(D, G) \\ &= \mathbb{E}_{m \sim p_m(m)}[\log D(a, m)] \\ &\quad + \mathbb{E}_{a \sim p_a(a)}[\log(1 - D(a, G(a)))] \end{aligned} \tag{6}$$

where D is discriminator, G is generator, a denotes aerial images, and m denotes map images. $\mathcal{L}_{\mathrm{adv}}$ trains the generator G and the discriminator D against each other. G tries to minimize this function, and D tries to maximize it.

TABLE I
COMPARISON OF COVERAGE AREA OF AIDOMG AND PP DATASET

Dataset	Image Sets	Regions	Continents	Longitude	Latitude
PP dataset	2,194	1	1	-74.0	40.7
AIDOMG	40,087	9	6	-115.2∼151.2	-33.9∼48.9

The perceptual loss [62] $\mathcal{L}_{\mathrm{per}}$ can improve visual quality of generated maps. The loss function is

$$\mathcal{L}_{\mathrm{per}} = \sum_{i=1}^{n} \frac{1}{N_i} \left\| F^{(i)}(m) - F^{(i)}(G(a)) \right\|_1 \tag{7}$$

where F is feature extraction network, $F_{(i)}$ denotes the first i layers of F and N_i is the number of pixels of corresponding feature map. We use a pretrained VGG model as a feature extraction network F.

In addition, we use features-matching loss [16] $\mathcal{L}_{\mathrm{fm}}$ to stabilize the training [16]. This loss penalizes differences between extracted features of real and generated images from multiple layers of the discriminator. The loss function is

$$\mathcal{L}_{\mathrm{fm}} = \sum_{i=1}^{n} \frac{1}{N_i} \left\| D^{(i)}(a, m) - D^{(i)}(a, G(a)) \right\|_1 \tag{8}$$

where $D^{(i)}$ denotes the first i layers of discriminator D, n denotes the total number of layers, and N_i denotes the number of elements in each layer.

Together, our loss for creative module is expressed as

$$\mathcal{L}_{\mathrm{CM}} = \lambda_{\mathrm{con}} \mathcal{L}_{\mathrm{con}} + \lambda_{\mathrm{adv}} \mathcal{L}_{\mathrm{adv}} + \lambda_{\mathrm{per}} \mathcal{L}_{\mathrm{per}} + \lambda_{\mathrm{fm}} \mathcal{L}_{\mathrm{fm}} \tag{9}$$

where we let $\lambda_{\mathrm{con}} = \lambda_{\mathrm{per}} = \lambda_{\mathrm{fm}} = 10$ and $\lambda_{\mathrm{adv}} = 1$ for loss balancing, following [16].

IV. AIDOMG DATASET

The dataset consisting of paired samples of aerial images and online maps is critical for developing and evaluating learning-based map generation methods. Thus, we collect and produce a large-scale dataset for online map generation called AIDOMG. It is collected from nine different areas of six different continents, including aerial images, original online map images, repainted online map images, and semantic segmentation labels. We first introduce the distinctive characteristics of AIDOMG and then elaborate on some details about it.

A. Characteristics

1) Large Scale: Scale and quality are necessary characteristics for a good dataset. We collect aerial images and corresponding online maps from nine different regions, and finally get 40 087 pairs of images. For comparison, as shown in Table I, the PP dataset [3], used by many online map generation methods [3]–[5], [17], has only 2194 pairs of images.

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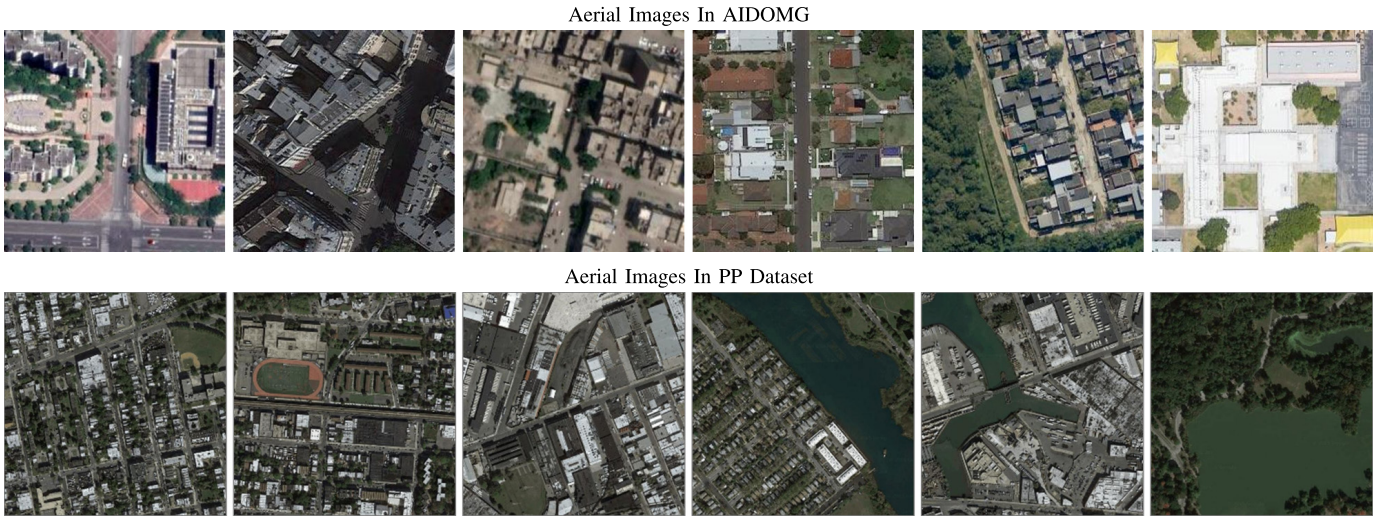


Fig. 3. Some examples of aerial images in PP dataset and AIDOMG. Compared with PP dataset collected from a single city, AIDOMG is diverse in terms of landform features, such as light, vegetation, and architectural style.

TABLE II
CROSS-REGION GENERATION PERFORMANCE OF PP DATASET AND AIDOMG IN WUHAN. USING A MULTICITY HYBRID AIDOMG DATASET CAN GENERATE BETTER MAPS IN UNFAMILIAR REGIONS

Model	Training dataset	Testing dataset	*FID* ↓	*1-NN score* ↓	*F1 score (road)* ↑	*F1 score (building)* ↑
pix2pixHD	PP dataset	AIDOMG-Wuhan	322.4343	1.0000	23.8738	38.9433
	AIDOMG (7 regions)	AIDOMG-Wuhan	**154.5909**	**0.9895**	**42.1929**	**52.7269**
DeepLabv3+	PP dataset	AIDOMG-Wuhan	232.2277	0.9986	31.2460	53.3037
	AIDOMG (7 regions)	AIDOMG-Wuhan	**213.7811**	**0.8959**	**50.1716**	**62.5451**
ours	PP dataset	AIDOMG-Wuhan	253.6267	1.0000	32.6118	52.2691
	AIDOMG (7 regions)	AIDOMG-Wuhan	**88.7803**	**0.8901**	**49.9016**	**60.6344**

2) Diverse Landform Features: Cities in different regions have different light conditions, vegetation types, and architectural styles, which together form the geomorphological features of aerial images. Map generation in a region different from the training set may get poor results, especially when the urban geomorphological features are widely disparate. We collect data from nine different regions of six different continents of the world so that the dataset contains more geomorphic features and the trained model can perform well on unfamiliar regions. As shown in Table I, our dataset is selected from a relatively wide range. Fig. 3 shows some examples of images from PP dataset and AIDOMG. Compared with the PP dataset collected from a single city, AIDOMG has diversity in terms of lighting, vegetation, and architectural style. In addition, we conduct experiments to compare the capability to generate maps on unfamiliar areas of the PP dataset with AIDOMG. Specifically, we train models on the PP dataset and AIDOMG, respectively, and generate online maps at regions, which are not covered by the training set. For the PP dataset, we use the training set divided in [3]. For AIDOMG, we mix the training set of seven regions (Haikou, Xining, Beijing, Las Vegas, Paris, Cairo, and Rio) to train the models. The above-mentioned two models will be tested on the data of the other two regions (Wuhan and Sydney), which are not used in the training set. To eliminate the impact of different mapping methods in the two datasets, we repaint online map images in the two datasets according to the same standard. Results in Tables II and III show that the model trained on multiple regions of AIDOMG has a significant advantage over the PP dataset when used in unfamiliar regions. Fig. 4 shows some qualitative results.

3) Semantic Information: In addition to aerial images and corresponding online maps, we also provide corresponding semantic segmentation labels for each aerial image. Semantic segmentation labels are obtained from online map images through unsupervised clustering including five categories: background, road, water, vegetation, and building. As shown in Fig. 5(e), these five semantic labels can basically cover the information expressed in online maps. Semantic labels can help neural networks to understand aerial images explicitly.

B. Details

1) Data Collection: The AIDOMG dataset is collected from Google Maps. We collect aerial images and online map images in nine regions, as shown in Table IV. All regions have the

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TABLE III

CROSS-REGION GENERATION PERFORMANCE OF PP DATASET AND AIDOMG IN SYDNEY. USING A MULTICITY HYBRID AIDOMG DATASET CAN GENERATE BETTER MAPS IN UNFAMILIAR REGIONS

Model	Training dataset	Testing dataset	*FID* ↓	*1-NN score* ↓	*F1 score (road)* ↑	*F1 score (building)* ↑
pix2pixHD	PP dataset	AIDOMG-Sydney	371.4082	1.0000	53.6534	56.7475
	AIDOMG (7 regions)	AIDOMG-Sydney	**236.0365**	**0.9912**	**80.5294**	**77.7998**
DeepLabv3+	PP dataset	AIDOMG-Sydney	303.5150	0.9997	62.9658	72.5790
	AIDOMG (7 regions)	AIDOMG-Sydney	**287.9500**	**0.9830**	**82.9339**	**83.6112**
ours	PP dataset	AIDOMG-Sydney	314.3452	1.0000	65.9953	75.7793
	AIDOMG (7 regions)	AIDOMG-Sydney	**191.9548**	**0.9697**	**82.9844**	**82.8188**



Fig. 4. Cross-region generation performance of PP dataset and AIDOMG (seven regions). The first two lines of pictures show the online images generated on Wuhan and the last two lines on Sydney. Using a multicity hybrid AIDOMG dataset can generate better maps in unfamiliar regions.

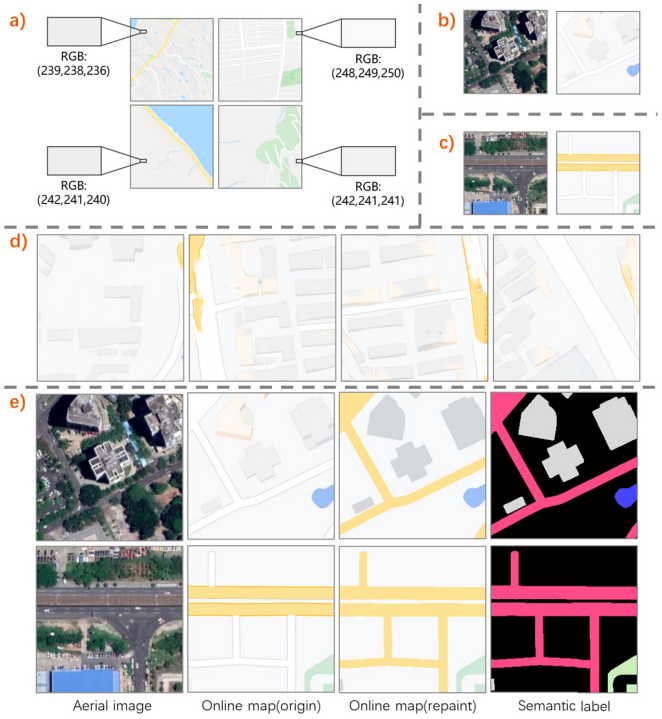


Fig. 5. Issues on online maps, and our processing. (a) Color values of the same elements in the original online map are slightly different. (b) Buildings are artificially divided into two colors, but there is no significant difference in aerial images. (c) Asphalt pavement has two representations in yellow and white on the online map. (d) Generated images often confused by color of roads and buildings. (e) Image comparison before and after processing.

same zoom level 18 with a spatial resolution of 0.6 m/pixel. For Haikou and Xining, we sampled from 79 × 45 and 36 × 62 tiles, respectively, to avoid capturing unmarked buildings in the countryside at the edge of the city. For other regions, we collected samples from 70 × 70 tile rectangles (about 100 km^2) to ensure that there are not many differences in the number of different regional datasets. Finally, 40 087 pairs of aerial images and online map images are collected.

2) Online Map Images Processing: We have noticed two issues on online maps that may affect the quality of the dataset. First, there is usually a slight difference in color value between different pixels representing the same element in an online map [Fig. 5(a)]. It provides no useful information for the map and may affect the training of the model. Second, some symbols in an online map have a strong artificially defined meaning. For example, in an online map, yellow is often used to indicate high-level highways, such as national highways and provincial highways, and white is used to indicate ordinary roads [Fig. 5(c)]. These differences are usually defined manually, and it's hard for people or neural network models to distinguish between them only with aerial images because their visual features of them do not differ much. We observe more significant confusion in our experiments, as shown in Fig. 5(d). Thus, we repaint online map images in our AIDOMG to eliminate these two problems and improve the quality of the dataset. Fig. 5(e) shows online map images before and after repainting. In addition, we perform unsupervised clustering on online map images using *k*-means algorithm, label clustering

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TABLE IV
DETAILS OF NINE PARTS OF AIDOMG. EACH PART IS FROM A DIFFERENT REGION AND THEREFORE, AIDOMG CAN COVER MORE DIVERSE FEATURES

City	Number of Image Sets	Train Samples	Test Samples	Zoom Level
Haikou(Asia)	3,555	2,488	1,067	18
Wuhan(Asia)	4,900	3,430	1,470	18
Xining(Asia)	2,232	1,562	670	18
Beijing(Asia)	4,900	3,430	1,470	18
Vegas(North America)	4,900	3,430	1,470	18
Paris(Europe)	4,900	3,430	1,470	18
Cairo(Africa)	4,900	3,430	1,470	18
Rio(South America)	4,900	3,430	1,470	18
Sydney(Australia)	4,900	3,430	1,470	18
Total	40,087	28,060	12,027	18

results and obtain semantic segmentation labels corresponding to each aerial image [Fig. 5(e)]. Overall, our dataset consists of aerial images, original online map images, repaint online map images, and semantic segmentation labels. Thus, our dataset can be used for both image translation and semantic segmentation.

V. EXPERIMENTS AND ANALYSIS

In this section, we first introduce details of experiments, such as datasets and metrics. Then, we evaluate and compare methods with extensive experiments. Finally, we conduct ablation studies to analyze our model.

A. Experimental Details

1) Metrics: To evaluate the visual quality of the generated online maps, we use Fréchet inception distance (FID) [63] and 1-nearest neighbor (1-NN) score [64] as metrics. FID measures the similarity of images at the feature level, which is closely related to the quality of human vision and is most often used to evaluate the quality of GAN's results [63]. It uses a well-pretrained Inception v3 model to extract the feature vector of images and compares the difference between the corresponding feature vectors. The 1-NN score can evaluate not only the quality of generated images but also the level of overfitting [65]. It computes the leave-one-out accuracy of a 1-NN classifier trained on true samples S_r and generated samples S_g. In addition, considering the specificity of the map generation task, we want the generated image to be not only visually similar to the real image but also accurate so that we can use it as a map. To evaluate the accuracy of the generated online maps, we use relaxed F1 score [66] to evaluate the accuracy of the generated maps following [48], [67], and [68]. The relaxed F1 score is the harmonic mean of the relaxed precision and recall. The relaxed precision is defined as the fraction of detected pixels that are within ρ pixels of a true pixel, while the relaxed recall is defined as the fraction of true pixels that are within ρ pixels of a detected pixel. We set ρ to 3 pixels as default. Because this metric is usually used for single-category classification problems, we calculate relaxed F1 score separately for the more intensive and important two labels, roads, and buildings.

2) Dataset: Existing map generation methods usually train on a single region, i.e., the training set and testing set belong to different parts of the same city. We mainly conduct model comparison experiments on cities Haikou and Paris separately, where Haikou is relatively complex and Paris is relatively simple. We select about 70% as training set and about 30% as testing set. For AIDOMG-Haikou, we select 2488 images for training and the remaining 1067 as a testing set. For AIDOMG-Paris, we select 3430 images for training and the remaining 1470 as a testing set. We distinguish the training set from the testing set by the rank ordering of the images to ensure that they are in two regions that do not overlap at all. Regarding the multiregion hybrid training, we will investigate it in an ablation study.

3) Experiment Setup: There are 200 epochs during training. We use Adam optimizer for the discriminative module and creative module and set $\beta_1 = 0.5$, $\beta_2 = 0.999$. For the discriminative module, we initialize the backbone part using the Xception model pretrained on ImageNet and randomly initialize the remaining parts. The initial learning rate of the backbone part is 7×10^{-5} and that of the remaining parts is 7×10^{-4}. For the creative module, we use random initialization, and the initial learning rate is 2×10^{-4}. We train our model in an end-to-end way. Specifically, in each iteration, we first perform a complete forward operation, then compute the losses for both modules, sum the two losses and perform back propagation to update the parameters. The loss for the discriminative module $\mathcal{L}_{\mathrm{DM}}$ will be used to update the parameters of the discriminative module only, while the loss of the creative module $\mathcal{L}_{\mathrm{CM}}$ will be used to update the parameters of both modules.

B. Performance and Comparison

1) Methods for Comparison: To show the effectiveness of our method, we compare our method with semantic segmentation models (DeepLabv3+ [7], DeepUNet [8], and BRRNet [11]) and image-to-image translation models (pix2pix [3], pix2pixHD [16], GeoGAN [4], and SMAPGAN [5]). Tables V and VI show quantitative results of each method. Compared with the structure used in the creative and discriminative modules, our model has a significant improvement, e.g., on AIDOMG-Haikou, our FID is 127.8159 (242.4001 for DeepLabv3+ and 183.4144 for pix2pixHD). In addition, our results are also better than other methods (FID is 198.2710 for pix2pix, 320.2838 for GeoGAN, 230.0857 for DeepUNet, and 302.3507 for BRRNet). SMAPGAN can achieve good results on PP datasets, but may not perform well on AIDOMG. Fig. 6 shows that online map images generated by our method are relatively better. The quality of online map images we generate is higher than that of other image-to-image translation methods, whereas the semantic segmentation methods generate images with significant problems of visual feature, e.g., roads are discontinuous or have rough edges.

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TABLE V
FID, 1-NN SCORE, AND F1 SCORE ON AIDOMG-HAIKOU. OUR MODEL SHOWS ADVANTAGES OVER THESE EXISTING METHODS

Model		*FID* ↓	*1-NN score* ↓	*F1 score (road)* ↑	*F1 score (building)* ↑
image-to-image translation	pix2pix	198.2710	0.9925	52.3275	67.7708
	pix2pixHD	183.4144	0.9906	62.5917	65.2247
	GeoGAN	320.2838	1.0000	40.9675	31.7118
	SMAPGAN	242.6285	0.9995	62.6069	68.8420
semantic segmentation	DeepLabv3+	242.4001	0.9911	69.2900	71.7900
	DeepUNet	230.0857	0.9883	45.2132	67.6547
	BRRNet	302.3507	1.0000	37.9612	64.6052
ours		**110.8562**	**0.9035**	**70.0497**	**71.8721**

TABLE VI
FID, 1-NN SCORE, AND F1 SCORE ON AIDOMG-PARIS. OUR MODEL SHOWS ADVANTAGES OVER THESE EXISTING METHODS

Model		*FID* ↓	*1-NN score* ↓	*F1 score (road)* ↑	*F1 score (building)* ↑
image-to-image translation	pix2pix	212.1710	0.9912	77.8321	85.9083
	pix2pixHD	189.3764	0.9840	77.1864	85.5952
	GeoGAN	287.4845	1.0000	66.3778	83.7189
	SMAPGAN	201.3030	0.9993	80.3709	90.3271
semantic segmentation	DeepLabv3+	228.0059	0.9939	82.4093	90.4086
	DeepUNet	224.8380	0.9976	70.4506	88.7658
	BRRNet	282.0488	0.9963	79.6973	90.4852
ours		**127.9603**	**0.9177**	**83.4667**	**90.9086**

TABLE VII
FID, 1-NN SCORE, AND F1 SCORE ON ORIGINAL ONLINE MAPS. OUR MODEL CAN STILL SHOWS ADVANTAGES

Online map	Model	*FID* ↓	*1-NN score* ↓	*F1 score (road)* ↑	*F1 score (building)* ↑
Original	pix2pix	276.1732	0.9972	7.8909	13.9074
	pix2pixHD	164.3329	0.9878	56.0085	61.7548
	ours	**103.1331**	**0.8964**	**65.4729**	**70.9311**
Repaint	pix2pix	198.2710	0.9925	52.3275	67.7708
	pix2pixHD	183.4144	0.9906	62.5917	65.2247
	ours	**110.8562**	**0.9035**	**70.0497**	**71.8721**

More specifically, we can compare the intermediate result (semantic image) generated by the combined method with the final result. As can be seen from Fig. 7, compared with the intermediate result, the final result has a similar structure, but the image quality is significantly improved.

C. Ablation Study

1) Training on Original Online Maps: Considering that the map repainting process may have some impact on online map data, we keep the original online map as an optional training data in the dataset. As in Table VII, the experimental results show that our method still retains the advantages when trained with original online maps. In addition, by comparing the performance of the same method trained on the original and repaint online maps, we can find that the repaint processing of the online map affects the visual performance of online maps, but can significantly improve the accuracy of the generated map.

2) Effect of Losses: Here, we verify the validity of the losses. As shown in Table VIII, we can see that the several losses used in our method have a positive impact on the final results, where the features-matching loss, focal loss, and Lovász loss have a significant effect on the visual quality and accuracy of the results. The content loss and perceptual loss also have a positive effect on visual quality and accuracy.

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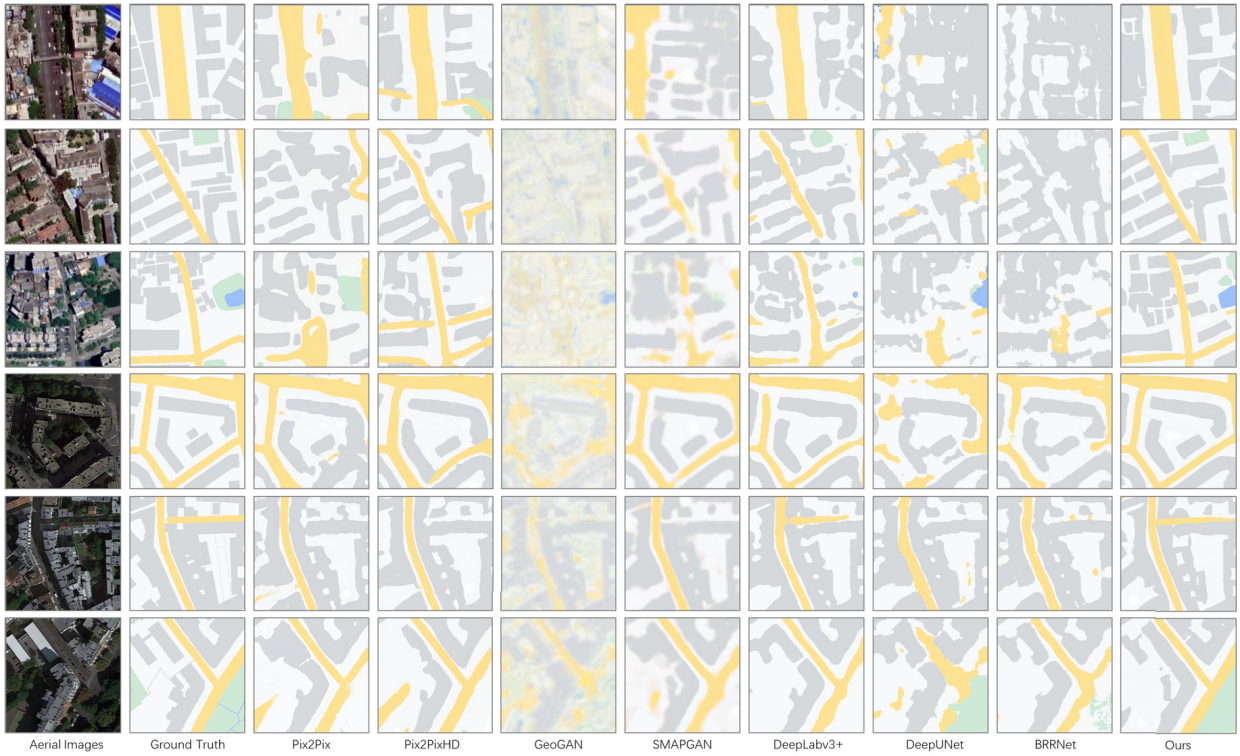


Fig. 6. Qualitative results of different methods on AIDOMG-Haikou and AIDOMG-Paris. Our results show more accurate identification of ground object elements and better road structure.

TABLE VIII
IMPACT OF USING DIFFERENT LOSSES. EACH LOSS HAS POSITIVE EFFECT ON VISUAL QUALITY AND ACCURACY

Model	*FID* ↓	*1-NN score* ↓	*F1 score (road)* ↑	*F1 score (building)* ↑
ours (without perceptual loss)	117.7685	0.9171	68.9895	71.6150
ours (without feature matching loss)	168.3496	0.9878	66.1372	69.1404
ours(without content loss)	116.2198	0.9114	69.0772	71.5963
ours(without focal loss)	159.9853	0.9681	68.7209	71.4690
ours(without lovász loss)	176.7862	0.9831	69.7080	70.7187
ours	**110.8562**	**0.9035**	**70.0497**	**71.8721**

TABLE IX
REPLACE THE FIRST MODULE AS OTHER AERIAL IMAGE INFORMATION EXTRACTION METHODS DO NOT LEAD TO LARGE ENHANCEMENTS

Model	*FID* ↓	*1-NN score* ↓	*F1 score (road)* ↑	*F1 score (building)* ↑
Pix2pixHD	183.4144	0.9906	62.5917	65.2247
Watershed & pix2pixHD	222.7492	0.9977	61.8347	65.4641
Edge detection & pix2pixHD	214.2561	0.9986	54.6377	62.7266
Region growing & pix2pixHD	222.5093	0.9981	52.1852	60.0814
Pix2pix & pix2pixHD	207.7539	0.9948	61.8494	65.5697
ours	**110.8562**	**0.9035**	**70.0497**	**71.8721**

3) Replace the First Module as Other Aerial Image Information Extraction Methods: In our approach, the discriminative module provides information to the creative module, which is responsible for the final generation. To test if semantic segmentation is the best choice, we have considered the existing aerial image information extraction methods as much as

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TABLE X

COMPARISON OF DIFFERENT MODULE STRUCTURES. OUR MODEL STILL MAINTAINS ITS ADVANTAGE WHEN THE NUMBER OF PARAMETERS IS CLOSE

Model		Parameter (in million)	*FID* ↓	*1-NN score* ↓	*F1 score (road)* ↑	*F1 score (building)* ↑
image-to-image translation	pix2pix (More Parameters)	242.98	216.2249	0.9995	30.0602	42.6204
	pix2pixHD (More Parameters)	235.90	228.0806	0.9986	43.4535	62.1107
	GeoGAN (More Parameters)	242.47	323.0896	1.0000	38.8553	51.8364
	SMAPGAN (More Parameters)	242.32	337.9447	0.9995	0.0005	43.9156
semantic segmentation	DeepLabv3+ (More Parameters)	237.07	272.3679	0.9995	61.4658	69.9317
	DeepUNet (More Parameters)	242.72	222.4467	0.9930	42.7063	47.6202
	BRRNet (More Parameters)	243.04	270.2349	0.9995	39.3765	60.4010
ours		243.51	**110.8562**	**0.9035**	**70.0497**	**71.8721**



Fig. 7. Intermediate semantic segmentation results and final online map results. We can see that the final result has basically the same element position as the intermediate result. At the same time, the final result has the characteristics that an online map should have. This shows that the generation module can effectively use the semantic information in the intermediate results and can learn the features of online maps from ground truth.

possible and conducted comprehensive experiments. Specifically, we select conventional segmentation methods (watershed algorithm, edge detection algorithm, and region growing algorithm) and image generation method (pix2pix [3]) to replace our discriminative module. The results shown in Table IX indicate that our semantic module outperforms other candidate methods by a large margin and achieves the best results in map generation. Semantic segmentation methods as the discriminative module can provide richer semantic information for each pixel, which is difficult for GAN to acquire on its own. Among the other alternative comparison methods, the traditional methods are heuristic and extracted features are not task-specific, whereas the features extracted by GAN are similar to those of the creative module, which cannot provide extra information. These make them inferior choices.

4) Effect of Number of Parameters: To compare methods with a similar number of model parameters, we keep a similar number of parameters for all models by increasing the number of channels per layer of networks. As shown in Table X, our method is still able to maintain its advantage when the number of parameters is close. It shows that our two-module network structure indeed improves the model performance, not because of the increase in the number of parameters.

5) Feature Space for Fusion: To investigate the impact of the combination method between the two modules, we combine the two modules on different feature spaces, i.e., prediction space, softmax space, and feature map space. The prediction is a 1-D label image, the softmax is a 5-D probability image (because there are five categories of semantic labels), and the feature map has 256 channels. The experimental results (Table XI) show that using either feature map space or softmax space is better than using prediction space. This suggests that the creative module does not simply use semantic labeling results provided by the discriminative module, but effectively uses richer information, such as confidence levels, to make more detailed judgments.

6) Model Structure Replacement: To test the generalization ability of our two-module approach, we replace the discriminative and creative modules with new structures, respectively. Specifically, we replace the discriminative module using semantic segmentation method OCRNet [30] and replace the creative module with image-to-image translation method

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TABLE XI

COMPARISON OF DIFFERENT FEATURE SPACE FOR FUSION. USING FEATURE MAP OR SOFTMAX CAN ACHIEVE BETTER PERFORMANCE THAN USING PREDICTION RESULT

Model	connection	*FID* ↓	*1-NN score* ↓	*F1 score (road)* ↑	*F1 score (building)* ↑
ours	feature map (base)	**110.8562**	**0.9035**	**70.0497**	**71.8721**
	softmax	159.9228	0.9667	69.6618	71.3341
	prediction	175.6430	0.9888	68.0437	71.2567

TABLE XII

COMPARISON OF DIFFERENT MODULE STRUCTURES. OUR MODEL IS COMPATIBLE WITH LATEST SEMANTIC SEGMENTATION OR IMAGE-TO-IMAGE TRANSLATION METHODS AND CAN IMPROVES PERFORMANCE BY USING THEM

Model		*FID* ↓	*1-NN score* ↓	*F1 score (road)* ↑	*F1 score (building)* ↑
image-to-image translation	pix2pix	198.2710	0.9925	52.3275	67.7708
	pix2pixHD	183.4144	0.9906	62.5917	65.2247
semantic segmentation	DeepLabv3+	242.4001	0.9911	69.2900	71.7900
	OCRNet	205.2270	0.9906	70.1067	73.0104
ours	DeepLabv3+ & pix2pix	149.7397	0.9719	69.5558	72.2235
	DeepLabv3+ & pix2pixHD (base)	**110.8562**	**0.9035**	70.0497	71.8721
	OCRNet & pix2pix	166.3538	0.9888	69.9548	73.2066
	OCRNet & pix2pixHD	165.8979	0.9892	**70.3443**	**73.2448**

TABLE XIII

COMPARISON OF DIFFERENT LOSS WEIGHTS OF CREATIVE MODULE AND DISCRIMINATIVE MODULE. OUR METHOD IS ROBUST TO DIFFERENT LOSS WEIGHT, BUT AN IMBALANCED LOSS WEIGHT WILL LEAD TO A SIGNIFICANT REDUCTION IN RESULTS

Model	λ (in Eq. 1)	*FID* ↓	*1-NN score* ↓	*F1 score (road)* ↑	*F1 score (building)* ↑
ours	0.005	156.4036	0.9588	69.3174	70.8166
	0.05	180.2967	0.9869	69.7376	71.4267
	0.25	129.7928	**0.9035**	68.2733	71.6357
	0.5	**110.8562**	**0.9035**	**70.0497**	**71.8721**
	1	129.3874	0.9452	68.7627	71.6685
	5	128.1802	0.9330	68.7719	71.2922
	50	161.3830	0.9794	69.1752	70.9014

pix2pix [3]. When using OCRNet, we combine two modules in softmax space for stabler training, since OCRNet has 512 channels of feature maps. Table XII shows that when we replace modules with other semantic segmentation methods and image-to-image methods, our model still works and can achieve better results compared with individual models. Our model is compatible with the latest semantic segmentation or image-to-image translation methods and achieves better performance than them.

7) Loss Weight Adjustment: We consider the influence of loss weight of discriminative module and creative module. The results in Table XIII show that our method is robust to different loss weights, but an imbalanced loss weight will lead to a significant reduction in results. This indicates that both the discriminative module and the creative module play an important role in our method.

8) Effect of Mixing Training Data From Different Areas in AIDOMG: We try to mix training data from different regions to explore whether they interfere with or enhance each other and whether mixed training has a boost on cross-domain generation. We first train and test the model on data of seven regions (Haikou, Xining, Beijing, Las Vegas, Paris, Cairo, and Rio) separately, i.e., we train seven models separately on the seven regions and test these models on their respective corresponding regions. Then, we train the model on the mixed data of seven regions. To verify the cross-regional performance of these models, we generate online maps and test on the other two regions (Wuhan and Sydney), respectively. For the generation of separately trained models on the two unused regions, we use each of the seven models to generate and calculate the average of the metrics. Rows 1–4 of Table XIV show that the model trained on mixing data

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TABLE XIV
USING AIDOMG SEPARATELY OR COMBINED FOR TRAINING. THE MODEL TRAINED ON COMBINED DATA PERFORMS BETTER WHEN GENERATING ON UNFAMILIAR REGIONS

Model	testing set	training mode	*FID* ↓	*1-NN score* ↓	*F1 score (road)* ↑	*F1 score (building)* ↑
ours	AIDOMG-Wuhan	separate	193.5488	0.9850	35.2395	41.8633
		combined	**88.7803**	**0.8901**	**49.9016**	**60.6344**
	AIDOMG-Sydney	separate	252.0986	0.9931	64.3308	68.2847
		combined	**191.9548**	**0.9697**	**82.9844**	**82.8188**
	AIDOMG (7 regions)	separate	78.5775	0.8973	**74.7867**	**79.4347**
		combined	**67.5756**	**0.8567**	72.9428	77.9770

performs significantly better when used on unfamiliar regions. Rows 5 and 6 show that the model trained on mixing data performs close to the performance of multiple models trained separately in each city when used on familiar regions.

VI. CONCLUSION

In this article, we propose an end-to-end method to automatically generate online maps from aerial images. This method uses a two-module structure to simulate the human behavior of drawing maps and achieve the best performance comparing with the previous state-of-the-art map generation methods [3]–[5] and semantic segmentation methods [7], [8], [11]. In addition, we collect and produce a large-scale aerial image dataset for online map generation called AIDOMG. It provides 40 087 samples of aerial images, online map images, and semantic labels collected from nine different regions of six continents, which can be used as the foundation of future researches on online map generation tasks.

In the future, we plan to collect and produce map-generation datasets with larger scales, more regions, richer ground features, and better label quality. We hope our preliminary work can serve as a foundation and push the field forward.

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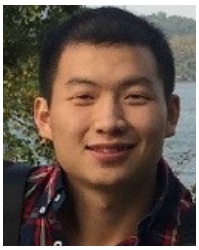
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